

time-consuming tasks for HDL languages users is coding test benches to verify the operation of their design. This is accomplished by representing each bus transaction graphically and then automatically generating the code for each transaction. TestBench makes use of the powerful features of the language that is being generated so that the engineer does not have to hand-code each transaction.

RapidSmith

The RapidSmith project is a set of tools and application programming interfaces (APIs) written in Java that aim to provide academia with an easy-to-use platform to try out experimental ideas and algorithms on modern Xilinx FPGAs. RapidSmith is based on Xilinx Design Language (XDL), which provides a readable file format equivalent to the Xilinx proprietary Netlist Circuit Description (NCD). With RapidSmith users can import XDL/NCD, manipulate, place, route and export designs among a variety of design transformations.

The RapidSmith project makes an excellent test bed to try out new ideas and algorithms for FPGA CAD research as code can quickly be written to take advantage of the APIs available. It also contains packages that can parse/export bit-streams (at the packet level) and represent frames and configuration blocks in the provided data structures. It can parse, manipulate and export bit-streams according to Xilinx documented methods.

RapidSmith creates special part files from device files created by ISE tools, which can then be used by RapidSmith for design manipulation. This project provides researchers the ability to leverage all of the XDL work previously done and avoid duplication. This will enable researchers to have more time to focus on what matters most—their research of new ideas and algorithms.

If you are using Eclipse as your IDE, it contains project files to get the project up and running with minimal effort. Just extract the files into your

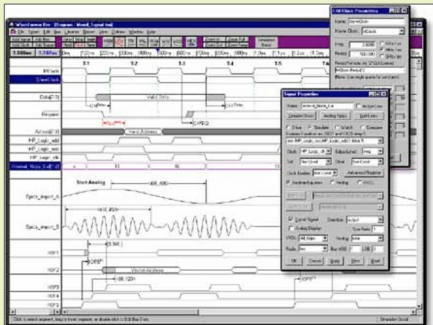


Fig. 1: Simulation under SynapticAD

workspace. Then, in Eclipse, you can import the project by going to 'File→Import,' then choose 'General/Existing Projects into Workspace.' On the next screen, choose 'Select root directory,' followed by the newly created folder 'rapidSmith.' Then click 'Finish.' This should load up the project in your workspace. You may have to restart Eclipse after setting the environment variables to get things to work.

Visual HDL

VisualHDL is an IDE for rapid design of FPGAs. It is based on THDL++, which is a C++ like language with fully parallel VHDL semantics. It allows you to develop fully parallel FPGA designs with the flexibility, modularity and maintainability of C++ code.

The IDE supports code completion, code navigation, design visualisation and several other useful features. The IDE is fully integrated with Xilinx ISE tool-chain. Moreover, a command-line THDL++ to VHDL compiler can be used in conjunction with any other tool that supports VHDL.

Using Visual HDL, instead of converting the inherently sequential C++ code into fully parallel VHDL synthesisable for FPGAs, users

can port the productivity of C++ and C++ IDEs to the FPGA world. THDL++ language provides a C++ like syntax and supports object-oriented features, templates, inheritance and reflection while being fully parallel by design.

THDL++ inherits VHDL semantics and provides several major time-saving features. It is a higher-level way to make VHDL designs without spending time on routine VHDL tasks.

SmartSim

SmartSim is a free and Open Source digital logic circuit design and simulation package. It lets you create complex circuits by allowing you to create your own custom components, and include them in other circuits as if they were any other built-in component. These larger circuits can then also be included in other designs as sub-components. SmartSim also offers the ability to print out or export your circuit designs to PDF, PNG or SVG format.

After designing the circuit, SmartSim offers an interactive simulation feature, allowing you to control your circuit and explore sub-components inside whilst the circuit is running. SmartSim also allows you to produce